Learner Brief — Personal Carbon Footprint & Savings Advisor

Project Goal

Build a **customer-facing sustainability product** that analyses spending behaviour, calculates carbon impact and provides practical recommendations to help customers reduce both **carbon emissions and costs**.

You will work as a professional technology team, applying software development, data, testing, security, DevSecOps and business skills.

There will be an award for the team delivering the best MVP at the final showcase.

Project background

Banks are increasingly focusing on sustainability and financial wellbeing. This prototype application will help customers understand the environmental impact of their spending and suggests cost-saving behavioural changes.

This project will help customers understand:

- how their spending impacts on the environment
- where they can they can reduce carbon emissions
- how they can save money through behavioural changes

The goal is to combine financial awareness with sustainability awareness.

Objectives

By the end of the project, your team will deliver a working MVP that allows a customer to:

- submit monthly spending data
- calculate estimated carbon impact
- receive category-based recommendations
- estimate the potential savings
- track progress over time
- complete sustainability challenges

Your MVP

Your product should allow a customer to:

- submit monthly spending

- calculate estimated carbon impact
- receive recommendations
- estimate potential savings
- view progress over time
- complete sustainability challenges

Technology

Choose **one** development language:

- **C# v8**
- **Python 3.14**

Your solution should use:

- GitHub for source control
- SQL for emissions and spending data
- automated testing
- local or Azure hosting
- GitHub Actions for CI/CD

Python 3.14

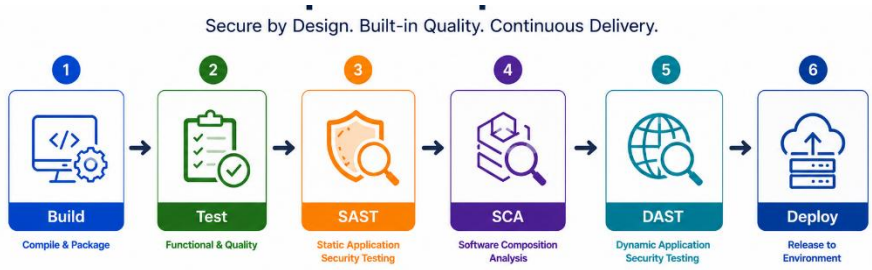
Recommended technologies:

- FastAPI
- Uvicorn
- pytest
- SQLAlchemy / SQL database connectivity

DevSecOps

Security is part of development throughout the project.

Your CI/CD pipeline should follow:



You should demonstrate:

- automated testing
- SAST
- dependency/SCA scanning
- DAST where practical
- secure coding
- input validation
- secure database access
- secrets management
- vulnerability remediation

Four Sprints of 3 days duration

Sprint 1 — Setup & First MVP

Outcome: Customer can submit spending and the data is securely stored.

Focus on:

- Agile planning
- GitHub and branching
- **C# v8 or Python 3.14**
- API
- SQL

- testing
- secure coding
- initial DevSecOps pipeline

Sprint 2 — Carbon Calculation

Outcome: Customer receives a carbon-impact estimate.

Focus on:

- emissions factors
- SQL/T-SQL
- carbon calculations
- data quality
- API development
- testing and security

Sprint 3 — Insights & Recommendations

Outcome: Customer understands their impact and receives recommendations.

Focus on:

- carbon insights
- trends
- category analysis
- recommendations
- potential savings
- dashboard development
- testing and DevSecOps

Sprint 4 — Final MVP & Showcase

Outcome: Customer-ready product.

Focus on:

- end-to-end testing
- security testing
- defect fixing

- deployment
- documentation
- DevSecOps evidence
- final presentation

Team & Professional Skills

Throughout the project, demonstrate:

- **Problem-solving**
- **Teamwork**
- **Cross-disciplinary collaboration**
- Communication
- Planning
- Stakeholder engagement
- Understanding and using individual strengths
- Commercial awareness
- Customer focus

Be prepared to explain **what you built, why you built it and the value it provides**.

Responsible AI

AI can be used to support:

- coding
- testing
- SQL
- troubleshooting
- documentation
- data analysis
- recommendations

However, you must **review, understand, test and security-check AI-generated output** before using it.

Stretch Objectives

Choose appropriate enhancements if the MVP is complete:

- AI-assisted recommendations
- predictive savings
- advanced dashboard analytics
- spending trends
- main branch protection
- authentication
- role-based access
- Azure Key Vault
- enhanced security controls

Final Showcase

Demonstrate:

- customer journey
- spending submission
- carbon calculation
- insights
- recommendations
- potential savings
- sustainability challenges
- dashboard
- automated tests
- CI/CD
- SAST, SCA and DAST
- security improvements
- deployment
- responsible AI use

Remember

MVP first. Build small. Test early. Secure throughout.

The strongest team will combine **technical capability, teamwork, communication, security and clear customer/business value**.